

of hardening, that all the default settings are changed, and everything that needs to be secured, is secure.

Now coming to the part about convincing the management, the open source databases we use are for internal efficiency. We have not yet exposed our external systems to databases that are open source. So, for Internet banking, I run Oracle. If I need high availability, I still use Oracle. But, if both these requirements are not there, why do I need to invest in these technologies? So, we have clearly demarcated how we will implement open source within the organisation.

**Q So, does that mean you don't use open source technologies when it comes to some of the very critical operations?**

Other than the operating systems (OS), we do not expose the rest to the Linux systems. Apart from the Red Hat Linux OS, none of our external technology is exposed to Linux.

**Q What is the reason for that? You don't have confidence in Linux technologies?**

No, it is not about the confidence or faith. It is more about the speed of support. For instance, if I am running Internet banking on Linux 24x7, where do I look for expertise in times of failure? That's the challenge we face. Moreover, there are not many people who have worked on those technologies. One of the key aspects that we look into is that the people we work with should be exposed to the technologies that we use. When we look at the risk matrix from the cost perspective, open source technologies give a benefit of 20-30 per cent. However, from the perspective of getting support, it becomes a little challenging. So that is why, wherever our need is critical, we haven't dabbled in open source.

**Q Can you elaborate on how the support bit becomes unmanageable with open source technologies?**

It is not about being difficult, it is about being challenging. For instance, in the case of open source, we have to identify a set of people who have worked on a certain open source technology. Whereas, in the case of Oracle, at any time I will find enough people who are exposed to this technology. So, the challenge is more in the resourcing part.

**Q So, I believe that the biggest advantage of open source technology is having a community around it. However, having only the 'community' is also its biggest disadvantage. Is this statement correct for a situation like yours?**

Yes, pretty much. There is always a response time required when it comes to harnessing the community benefits. If I have the luxury of time on my hands, where I can wait for the responses to come in, this model is okay. Normally, one is able to find a solution, but the problem comes up when one is unable to find a solution. The community in the open source world is extremely vibrant and there are people constantly working on the product, but you do not always get a response

when you need it. It may take some time. So, for critical functions for which every minute counts, we would prefer a solution that is available easily.

**Q When did you begin your tryst with open source technologies?**

We began our open source journey in 2009. We started looking at the Linux operating system, which some of our users could migrate to from proprietary operating systems. From the security perspective, Linux was pretty secure and regards the patching, it was pretty manageable. Also, from the performance perspective, it was superior. We benchmarked the Linux OS against what we used and we found that it is an appropriate option with respect to availability and need. At one point of time, about 60 per cent of our boxes were running on Linux in our data centre. After that, we started to diverge into various other open source technologies, like the MySQL database, load balancer, Alfresco, et al.

We also use OpenOffice software in our branches. So, if I am setting up a branch, I would prefer that a majority of people work on OpenOffice and only a couple of people work on Microsoft Office.

**Q So, you have started pushing open source technology at the user level as well?**

Yes. Like I said, 70 per cent of our staff in the ING Vysya branches were using OpenOffice.

**Q Has this changed after Kotak's acquisition?**

We are still in the integration phase. Kotak works with Microsoft Office all the way. We have not yet started deploying Microsoft Office all the way, in our branches. We are still deploying the same technologies that we used to, earlier.

**Q What problems did you face while deploying open source technologies in your environment?**

From the users in the firm, there is a lot of resistance. Some people tend to think that if they are given OpenOffice to work on, they are not important enough. So, we had to change that mindset. We had to educate users about the benefits of the software, the money it saves, and make them feel that they were contributing to the growth of the branch. Where central deployments were concerned, there were no challenges. We just had to ensure that there was enough support available if there was a glitch. In this case, the end users do not even know that they are working on open source technologies, as they use their Web page access to get into the system. For them, it doesn't matter what technology has been deployed at the back end, as long as the performance is fine and they get their work done. But as I said, there were several challenges we had to face while deploying open source technology at the desktop level. We also tried a pilot for deploying Linux on workstations, but it didn't work and we had to abandon that plan. 